UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Computer generations and history

Question 1 is based on the following snippet garnered from a computer science web page.

Henry Edward "Ed" Roberts
(September 13, 1941 -April 1, 2010)
was an American engineer, entrepreneur and medical doctor who invented the first commercially successful personal computer in 1975. He is most often known as "the father of the personal computer". The computer was available as a kit for $439 or assembled for $621 and had several additional add-ons such as a memory board and interface boards. By August 1975, over 5,000 Altair 8800 personal computers were sold; starting the personal computer revolution.

Figure 1: Snippet from a computer science web page.

a) In which computer generation that a personal computer is introduced?
b) Discuss ONE (1) other major discovery that is found in this generation.
c) Advances in laptop and hand-held devices are accelerating during this computer generation. How such technology is feasible as compared to previous computer generation?

Source: CSC429 (8).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Data representation and number systems

Calculate 86 - (-30) using the sign-and-magnitude method. Assume an 8-bit byte. Show your complete workflow.

Answer: left 86 -> 0 1010110; right -30 -> 1 0011110; 86 - (-30) = 116 -> sign-magnitude 0 1110100

- *1 mark for converting both operands to sign-magnitude.*
- *1 mark for applying the sign rule for subtraction.*
- *1 mark for the correct magnitude operation.*
- *1 mark for the correct sign bit.*
- *1 mark for the final decimal and byte representation.*

QUESTION A3 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.

Instruction
Opcode
Address A
Memory
Operand

Figure 2

Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source

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paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Parallel and multiprocessor architecture

Complete the following taxonomy of parallel processor architecture with appropriate example.

Single Instruction Multiple
Data Stream
ii
Processor Organization
Multiple Instruction Multiple
Data Stream
v

Source: CSC429 (5).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Input/output and interrupts

Keyboard is the most common means for computer interaction.
a) Explain TWO (2) types of characters that keyboard provides.
b) Describe THREE (3) categories of input and output module. Give an appropriate example for each category.

Source: CSC429 (3).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Input/output and interrupts

a)
Define interrupt.
b)
Differentiate between interrupt and direct manipulation access.
c)
Given the following scenario.
Dilan is surfing woolworth.com to buy groceries online. While he is comparing prices of products in woolworth.com and one other website, two messages popped up onto his screen. Figure 2 (a) and (b) illustrate the conveyed messages.

Are you sure you want to
There is a hardware event failure
in this computer. Please reboot
cancel this request?
this computer now.
(a)
(b)

Figure 2: The pop-up messages

i)
Identify the type of interrupt in Figure 2(a) and Figure 2(b) with proper justification.
ii)
While interruptroutinefor Interrupt Ais executing, it is interrupted by interrupt Bwhich has a higher priority than interrupt A. What will happen to interrupt routine A?

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A7 (10 marks)

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Topic: Computer architecture and CPU organisation

a) Describe the function of each computer bus type.
b) Discuss TWO (2) benefits of synchronous bus timing.

Source: CSC429 (7).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Memory hierarchy and cache

Question 7 is based on the following scenarios.
While cleaning his house, Ahmad found an old cassette tape of a popular 80's band. He played the tape and listened to several songs according to the listed tracks.
Hafiz was tasked to research on the history of Malay culture by his professor. He went

to the National Archive to search for information and manage to access one the resources which was stored in a floppy disk.
Sara is planning to buy a new computer. She aims to get one with good processing speed. While researching for computer specifications, she found out that larger cache space has a direct effect towards processing performance.
Hana loves playing role-playing games (RPG) and is planning to buy a newly released game that requires a high-speed RAM. But, she is devastated when she found out that her game console does not support the minimum RAM requirement of the game.
a)
Based on the above scenarios, identify the access method for each storage devices/memory mentioned in (i), (ii), (iii) and (iv).
b)
Describe SRAM and DRAM.
c) Which level of cache is the fastest in a three-level cache architecture? Which type of RAM is used to implement the cache?

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Computer architecture and CPU organisation

a) Discuss the function of each Central Processing Unit (CPU) component'
b) Elaborate on what will happen in the machine cycle if the bus system is faulty

Source: CSC429 (20).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Data representation and number systems

Answer both parts. Show your complete workflow.
a) Convert 256610 to hexadecimal.
b) Convert 7E816 to decimal.

a) Convert 256610 to hexadecimal.

Answer: 256610 = A0616

- *1 mark for repeated division by 16.*
- *1 mark for correct remainders.*
- *1 mark for reading remainders in reverse order.*
- *1 mark for the correct hexadecimal digits.*
- *1 mark for the base subscript.*

b) Convert 7E816 to decimal.

Answer: 7E816 = 202410

- *1 mark for place values 16^2, 16^1 and 16^0.*

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- *1 mark for correct digit values.*
- *1 mark for correct products.*
- *1 mark for correct addition.*
- *1 mark for the final decimal value.*

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 2 and Table 1.
y = A(A + 2B/C-D 2)
Figure 2
Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 1 using zero-, one-, and three-address instructions. The instructions available for use are provided in Table 1. Register A, B, C, and D cannot be modified, and registers x, y, and z can be used as temporary registers. However, the final answer must be stored in Register y.
Table 1: Machine Instruction
One-address
Three-address
Zero-address
PUSHp
LOADp
ADD (p,q,r)
POPp
STORp
SUB (p,q,r)
ADD
ADDp
MUL (p.q.r)
SUB
SUBp
DIV(p,q,r)
MUL
MULp
DIV
DIVp

Source: CSC429 (17).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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